

Problem

Cystic fibrosis patients have a mutation on their cystic fibrosis transmembrane conductance regulators, which are categorized into five classes. Each class has a different faulty part of the protein. The two most common are class II (delF508) and class III (g551d). The mutations prevent the flux of chloride ions through the membrane of the cell, either by preventing the protein from reaching the surface, or allowing it to reach the surface but not properly function.

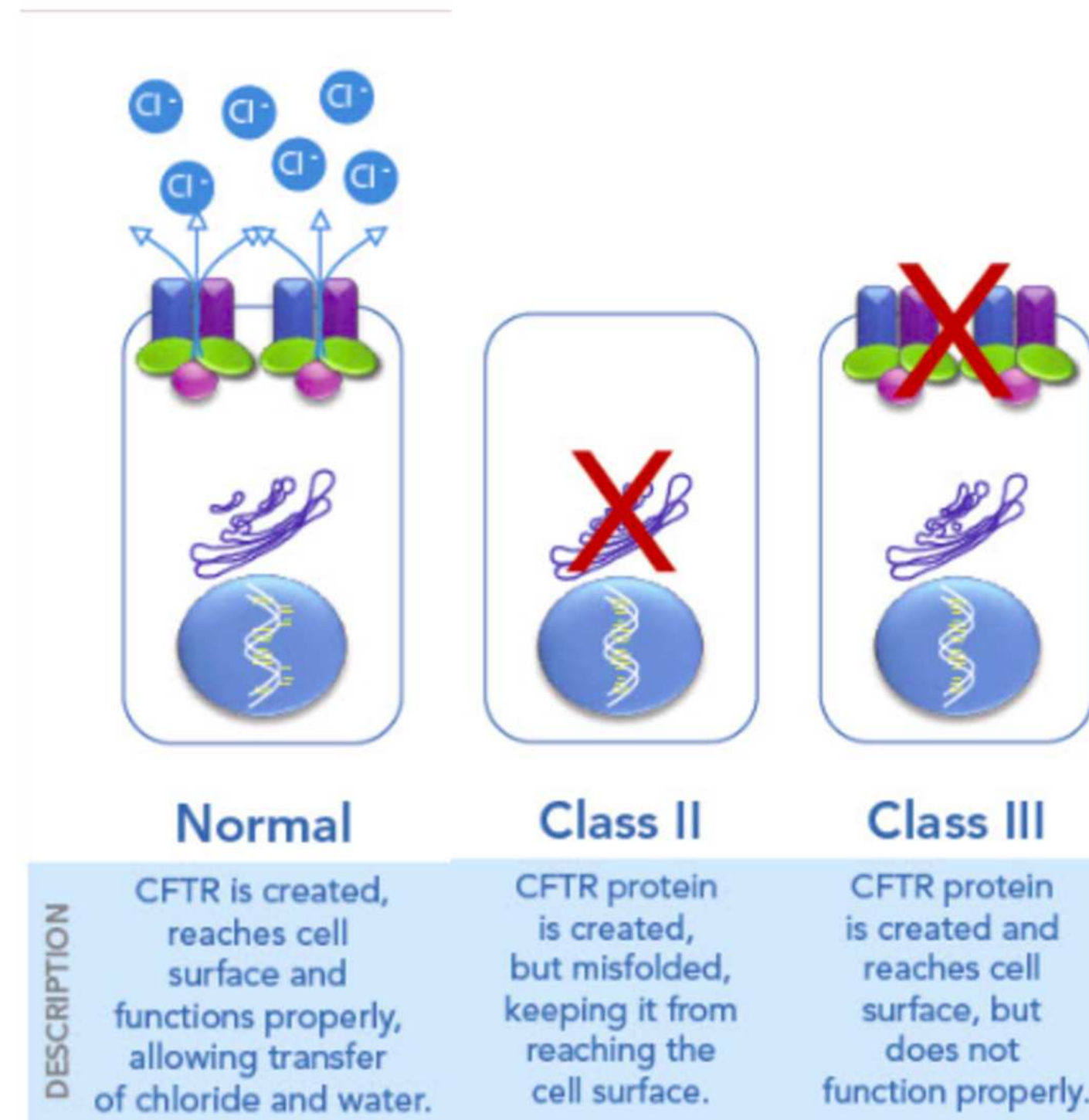
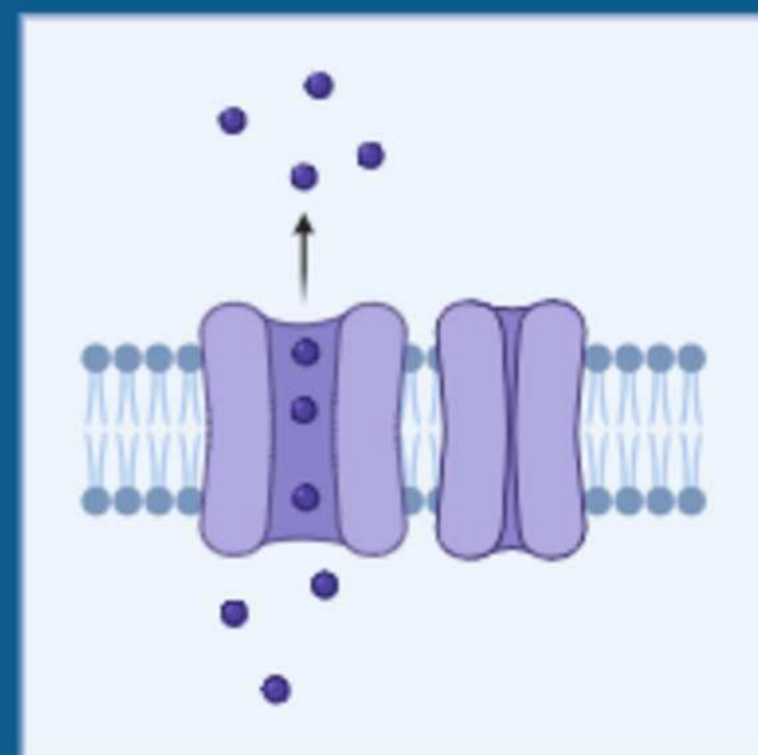


Figure 1 displaying the difference between the class II and class III CFTR mutations.

Background

CYSTIC FIBROSIS

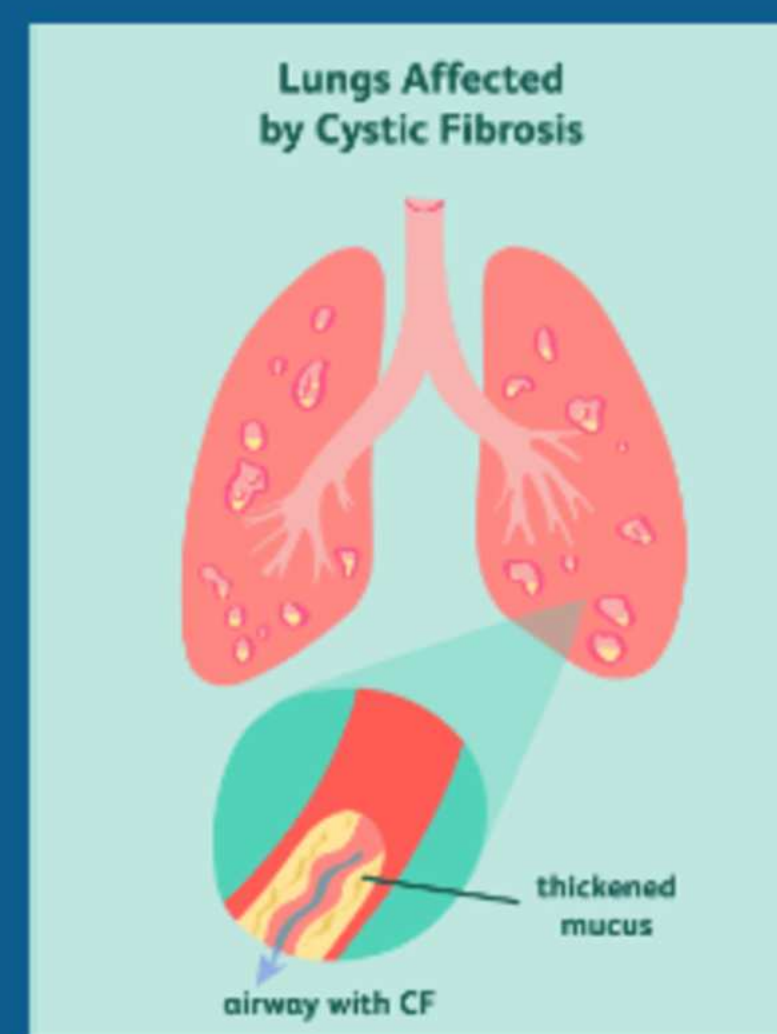
Genetic disease
→ Produces thick, sticky mucus
→ Builds up and clogs ducts and airways
→ Respiratory infections, difficulty breathing and sweating, and inability to absorb nutrients
→ Potential pancreatic insufficiency



Caused by mutation in the cystic fibrosis transmembrane conductance regulator (CFTR)



CF affects 1 in 2,500-3,500 newborns each year.



Most common in caucasian newborns

Currently incurable

Therapy treatments cost up to \$300,000

Created in BioRender.com

Figure 2 displaying background information on Cystic Fibrosis. Diagram created in BioRender.com

Hypothesis

The more copper supplemented into the cells, the higher the rate of chloride ion flux will be because copper is a trace element that many cystic fibrosis patients lack. Nutrition deficiency is very common among CF patients due to nutrient malabsorption and pancreatic insufficiency, caused by the buildup of mucus.

IV: Dosage of copper supplements (0mM, 0.25mM, 0.5mM, and 1.0mM).

DV: Proper functioning of the protein increase in chloride ion flux.

Control (-): Chloride ion flux of a wildtype (non-mutated) CFTR protein.

Increasing Chloride Ion Flux in Cystic Fibrosis Using Copper Supplements

Methods

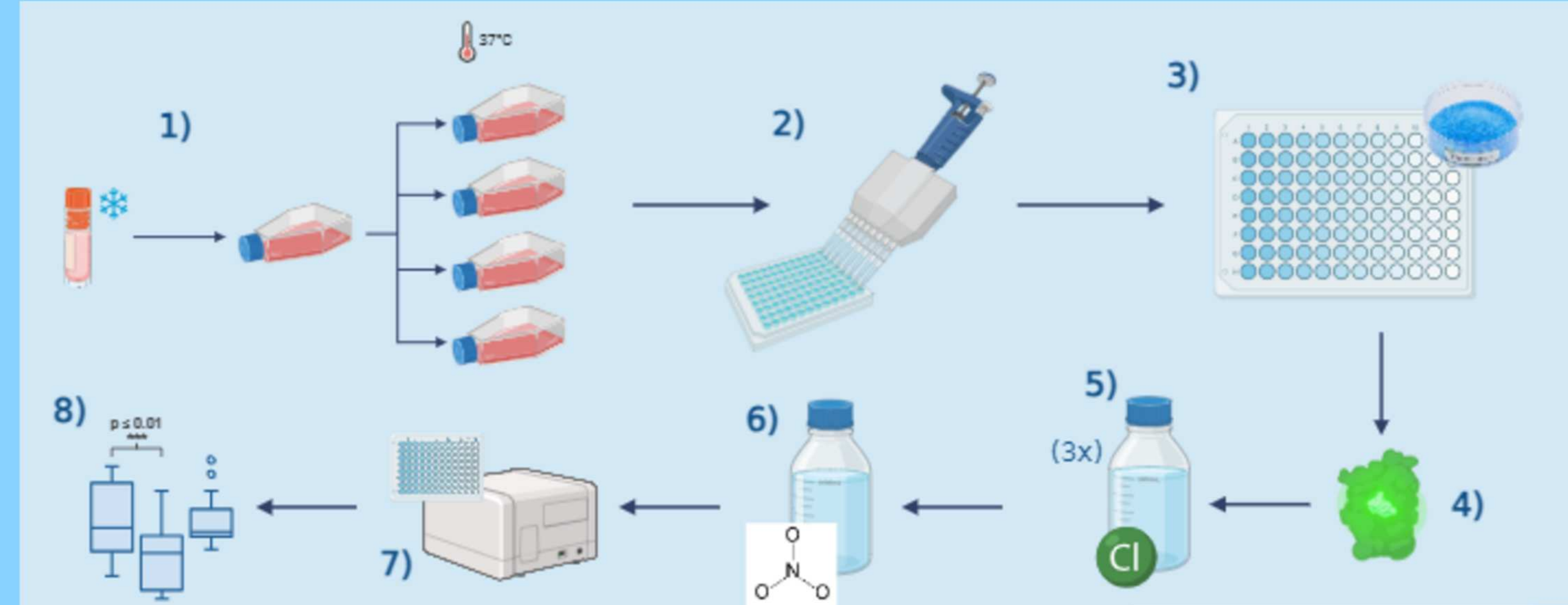
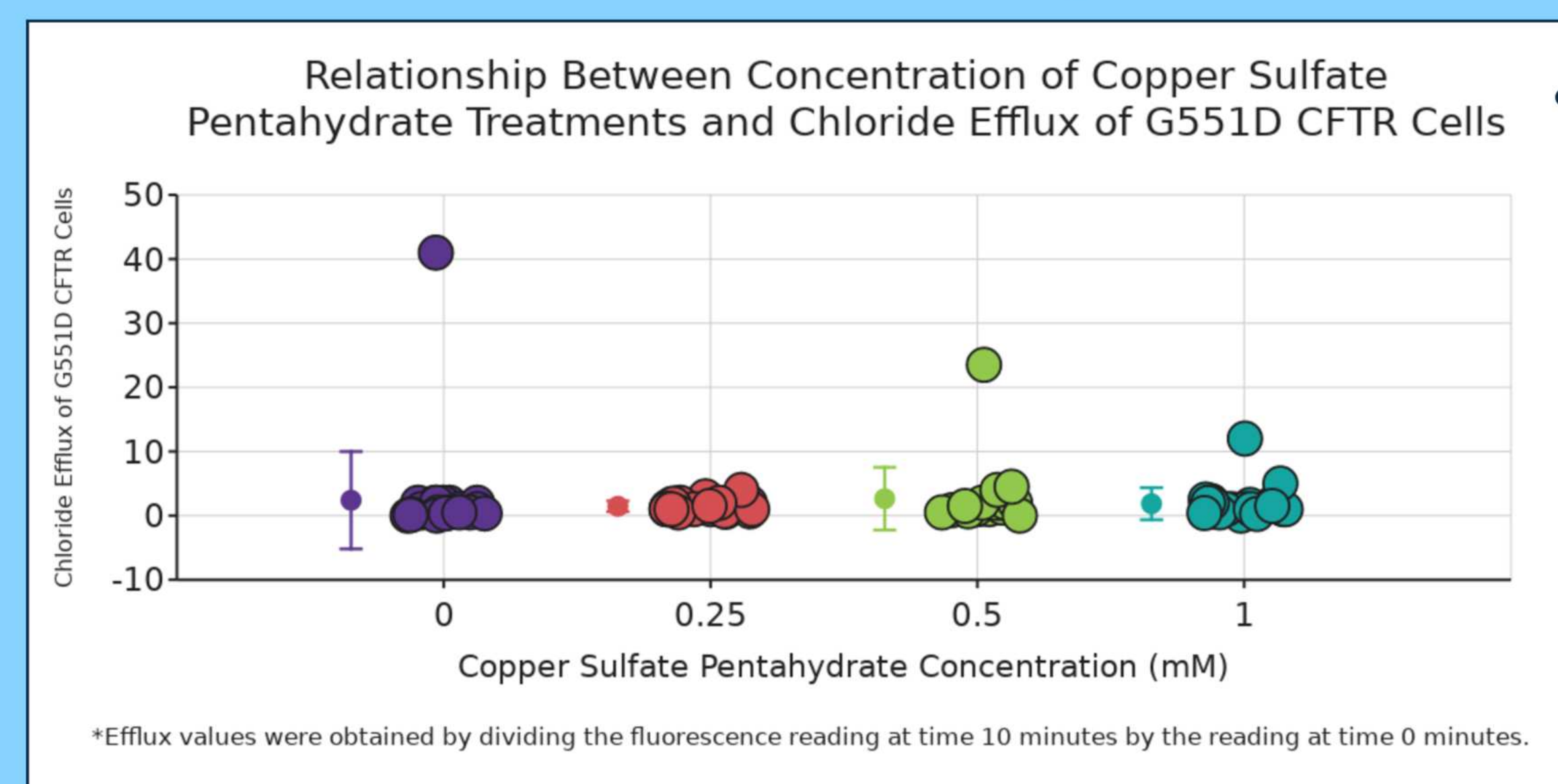


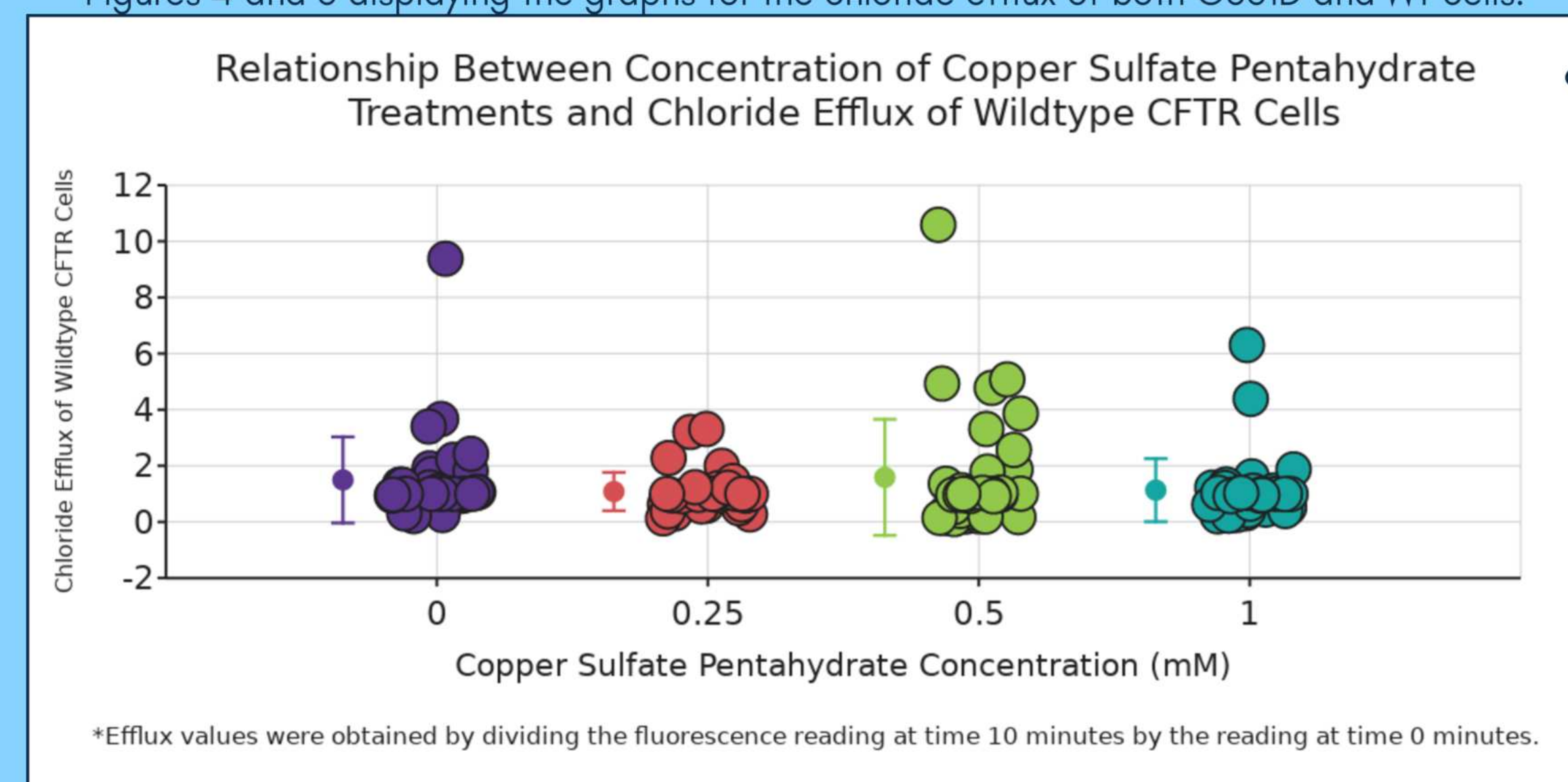
Figure 3 displaying the procedure for collecting data. Diagram created in BioRender.com

- 1) Thaw and subculture FRT cells (G551D and WT).
- 2) Seed black 96 well plate with 150µl per well of cell/growth media mixture.
- 3) Add copper treatments (16.5µl of 0mM, 0.25mM, 0.5mM, and 1.0mM). Let sit for 48 hours.
- 4) Soak with 150µl of 10mM SPQ (6-Methoxy-N-(3-Sulfopropyl)Quinolinium) for a few hours.
- 5) Rinse 3x with 150µl ringers solution (containing chloride).
- 6) Add 150µl ringers solution (containing nitrate).
- 7) Run fluorescence assay at 0 minutes and 10 minutes.
- 8) Kruskal wallis to test for statistical significance.
- 9) Compare the data between the control cell line and the mutated cell line to determine the optimal level of copper treatments.

Results



Figures 4 and 5 displaying the graphs for the chloride efflux of both G551D and WT cells.



Results (Continued)

Fluorescence readings obtained by conducting the fluorescence assay at time 0 minutes and 10 minutes were converted into chloride efflux values by dividing the time 10 fluorescence values by the time 0 fluorescence values. The p-values obtained for both cell types do not confirm that the differences between groups are statistically significant. however, a pattern was observed with the G551D cells, which seemed to have an optimal level of chloride efflux with 0.5mM of copper treatment.

Conclusion

- While not statistically significant since the p-values were greater than 0.05, the data in both cell lines suggest that 0.5mM of copper treatment produced the highest levels of chloride efflux, supporting the hypothesis which predicted that copper has some sort of role in the chloride efflux.
- Additionally, low values of chloride efflux were observed for 1.0mM of copper treatment, which could potentially suggest that too much copper could have a negative effect as well.
- Even though the statistics demonstrated that the difference between groups could have possibly been due to random variation, the copper did in fact contribute to small differences in the chloride efflux, even if the pattern was not always consistent.

Future work

- Since the p-values of 0.14 and 0.16 were not very high, and there is some indication that the results between copper concentrations were different, it may be worth it to continue the study by continuing to add more concentrations and more trials to solidify the results.
- One of the limitations of this research was the number of trials completed due to a late start on data collection.
- The class II delF508 mutation was never tested as hoped, so it would be a viable option to continue testing copper treatments in the future.
- While the effectiveness still remains unknown, copper supplements could be tested outside the lab with CF patients, whose chloride efflux could be characterized through the sweat test.

References

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